Milestone 4: Final Investment Memo Grading Rubric

QM 2023 Capstone Project

Total Points: 50 **Components:** Team Memo (40 points) + Individual Addendum (10 points)

Team Memo Grading (40 points)

1. Reproducibility & Technical Rigor (10 points)

Excellent (9-10 points):

- ☑ **Code runs end-to-end:** All scripts from M1, M2, M3 execute without errors
- ☑ **Results match memo:** Tables and figures in memo can be regenerated from code
- ☑ **Models are correct:** FE and DiD/ARIMA/ML specifications are sound
- ☑ **Diagnostics complete:** All assumption tests run; violations addressed
- ☑ **Robustness checks:** At least 3 checks conducted and results compared

Satisfactory (6-8 points):

- ☑ Code runs with minor warnings (not critical errors)
- ☑ Most results match, but 1-2 tables/figures have minor discrepancies
- ☑ Models mostly correct, but 1-2 specification issues
- ☑ Diagnostics run but not all violations addressed

Needs Improvement (3-5 points):

- ⚠ Code requires manual debugging to run
- ⚠ Significant discrepancies between code output and memo tables
- ⚠ Model specifications have errors
- ⚠ Minimal diagnostics or robustness checks

Unsatisfactory (0-2 points):

- ❌ Code does not run at all
- ❌ Results in memo cannot be reproduced
- ❌ Models are fundamentally flawed

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2. Structure & Clarity (10 points)

Excellent (9-10 points):

- ☑ **All required sections present:**
  - Executive Summary (0.5 page)
  - Methodology (1 page)
  - Results (1.5-2 pages)
  - Conclusions & Recommendations (1 page)

    - References (0.5 page)
    - AI Audit Appendix (0.5-1 page)
- ☑ **Length:** 5-7 pages (not too short, not too long)
- ☑ **Jargon-free:** Written for non-technical Investment Committee audience
- ☑ **Flow:** Logical progression from problem → methods → results → recommendations
- ☑ **Professional tone:** No typos, grammatical errors, or informal language

Satisfactory (6-8 points):

- ☑ Most sections present, but 1-2 are incomplete or rushed
- ☑ Length acceptable (4-8 pages)
- ☑ Some jargon remains but mostly accessible
- ☑ Flow is adequate but could be smoother
- ☑ Minor typos or grammatical errors

Needs Improvement (3-5 points):

- ⚠ Missing 2+ required sections
- ⚠ Too short (<4 pages) or too long (>10 pages)
- ⚠ Heavy jargon (unintelligible to non-economist)
- ⚠ Poor flow (sections feel disconnected)
- ⚠ Multiple typos or grammatical errors

Unsatisfactory (0-2 points):

- ❌ Missing most sections or memo is < 3 pages
- ❌ Completely inaccessible to non-technical audience
- ❌ Unprofessional (sloppy writing, formatting errors)

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3. Results & Interpretation (12 points)

Tables & Figures (6 points):

Excellent (6 points):

- ☑ **Table 1 (FE regression):** Publication-ready (columns: Variable, Coefficient, SE, t-stat, p-value, significance stars)
- ☑ **Table 2 (Model B):** Formatted similarly; side-by-side comparison if applicable
- ☑ **Figure 1:** Key visualization (e.g., dual-axis plot, sector sensitivity chart) with clear title, labels, legend
- ☑ **Figure 2:** Diagnostic plot (residuals vs. fitted or Q-Q plot) with caption
- ☑ **High resolution:** Figures are 300 DPI or higher (not pixelated)
- ☑ **Captions:** All tables/figures have captions explaining what they show

Satisfactory (4-5 points):

- ☑ Tables and figures present but formatting could be improved
- ☑ Some labels/legends missing
- ☑ Resolution adequate but not optimal
- ☑ Captions present but generic

Needs Improvement (2-3 points):

- ⚠ Tables are raw Python output (not formatted)
- ⚠ Figures are unlabeled or unclear
- ⚠ Low resolution (pixelated)
- ⚠ Missing captions

Unsatisfactory (0-1 points):

- ❌ No tables or figures
- ❌ Tables/figures are unreadable

Interpretation (6 points):

Excellent (6 points):

- ☑ **Coefficient interpretation:** "A 1 pp increase in FEDFUNDS reduces REIT returns by 2.5 pp" (economically clear)
- ☑ **Magnitude assessed:** "During 2022-2023 rate hike (5.25 pp), model predicts 13% decline"
- ☑ **Economic mechanisms explained:** Leverage channel, discount rate channel, demand effects
- ☑ **Robustness discussed:** "Results hold across alternative lags, COVID exclusion, sector subsamples"
- ☑ **Non-significant results acknowledged:** "SIZE is not significant ($p = 0.465$) after controlling for factors"

Satisfactory (4-5 points):

- ☑ Coefficients interpreted but not always in economic terms
- ☑ Magnitude mentioned but not assessed for importance
- ☑ Some economic reasoning but not comprehensive
- ☑ Robustness mentioned briefly

Needs Improvement (2-3 points):

- ⚠ Minimal interpretation ("Coefficient is negative and significant")
- ⚠ No discussion of magnitude or economic mechanism
- ⚠ No robustness discussion

Unsatisfactory (0-1 points):

- ❌ No interpretation provided
- ❌ Interpretation is incorrect

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4. Recommendations & Caveats (8 points)

Investment Recommendations (4 points):

Excellent (4 points):

- ☑ **Specific recommendations:** "Overweight Industrial REITs by 15%; neutral-to-underweight Retail/Office"

- ☑ **Grounded in evidence:** Recommendations follow directly from regression results
- ☑ **Scenario analysis:** "If Fed cuts 100 bp, expect 2.5% REIT recovery"
- ☑ **Risk-adjusted:** Acknowledges uncertainty and market conditions

Satisfactory (3 points):

- ☑ Recommendations provided but vague ("Be cautious with REITs")
- ☑ Some connection to results but not explicit
- ☑ Scenario analysis mentioned but not quantified

Needs Improvement (1-2 points):

- ⚠ Generic recommendations ("Diversify across sectors")
- ⚠ No clear connection to empirical findings

Unsatisfactory (0 points):

- ❌ No recommendations provided

Caveats & Limitations (4 points):

Excellent (4 points):

- ☑ **Honest acknowledgment:** "Our FE model assumes time-invariant REIT characteristics, which may be violated if portfolios restructure"
- ☑ **Model-specific:** Discusses parallel trends (DiD), omitted variables (FE), stationarity (ARIMA), or interpretability (ML)
- ☑ **Implications:** "If parallel trends are violated, our DiD estimate may overstate the causal effect"
- ☑ **Mitigation suggested:** "Future work could test robustness with sector-time interactions"

Satisfactory (3 points):

- ☑ Limitations mentioned but not deeply discussed
- ☑ Generic caveats ("Data limitations, external validity")

Needs Improvement (1-2 points):

- ⚠ Minimal caveats ("Our analysis has some limitations")
- ⚠ No model-specific discussion

Unsatisfactory (0 points):

- ❌ No limitations acknowledged
- ❌ Claims results are definitive with no caveats

Individual Addendum Grading (10 points)

Graded per student (not team average)

1. Specific Contribution (3 points)

Excellent (3 points):

- ☑ **Detailed task list:** Each milestone has 2-3 specific tasks (e.g., "Implemented FRED API, 20 hours")
- ☑ **Hours estimated:** Realistic time estimates for each task
- ☑ **Key deliverables:** Identifies specific sections of code or memo authored

Satisfactory (2 points):

- ☑ Tasks listed but not highly specific ("Helped with M1")
- ☑ Hours not estimated or vague

Needs Improvement (1 point):

- ⚠ Vague contribution ("I helped with coding and analysis")

Unsatisfactory (0 points):

- ✕ No contribution details provided

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2. Defended Decision (3 points)

Excellent (3 points):

- ☑ **Clear decision:** "I recommended using a 2-month lag for FEDFUNDS"
- ☑ **Evidence-based reasoning:** "M2 lag analysis showed strongest correlation at lag 2; economically sensible (REITs negotiate leases over 1-2 months)"
- ☑ **Robustness:** "M3 robustness checks confirmed lag 2 is most statistically significant"
- ☑ **Alternative considered:** "We considered lag 0 but rejected it due to weak significance"

Satisfactory (2 points):

- ☑ Decision stated and reasoning provided, but not deeply grounded in evidence
- ☑ No alternatives discussed

Needs Improvement (1 point):

- ⚠ Decision stated but reasoning is weak or generic

Unsatisfactory (0 points):

- ✕ No defended decision provided

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3. Key Limitation (3 points)

Excellent (3 points):

- ☑ **Substantive limitation:** "Our FE model assumes time-invariant REIT characteristics, but COVID-era portfolio restructuring may violate this"
- ☑ **Why it matters:** "If REITs shifted to industrial properties in 2020, they'd appear less rate-sensitive, but this could be due to portfolio change, not rate environment"
- ☑ **Mitigation suggested:** "Two-way FE with sector-time interactions could partially address this"

Satisfactory (2 points):

- ☑ Limitation stated but not deeply discussed
- ☑ Implications mentioned but not explained

Needs Improvement (1 point):

- ⚠ Trivial limitation ("Sample size could be larger")

Unsatisfactory (0 points):

- ❌ No limitation discussed

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4. Overall Quality (1 point)

Excellent (1 point):

- ☑ Professional writing, no typos
- ☑ Honest reflection (acknowledges both strengths and weaknesses)
- ☑ Exactly 1 page (not 0.5 pages, not 2 pages)

Unsatisfactory (0 points):

- ❌ Poorly written or dishonest
- ❌ Wrong length (<0.75 pages or >1.25 pages)

AI Audit Appendix Requirement (Pass/Fail, Zero Credit if Missing)

Required for Team Memo: `AI_AUDIT_APPENDIX.md` as part of memo or in Appendix section

Pass (Credit for M4):

- ☑ Summary of AI use across all milestones (M1, M2, M3, M4)
- ☑ Key verification and critique examples
- ☑ Responsibility statement ("We take full ownership of all work")

Fail (Zero Credit for M4):

- ❌ No AI Audit Appendix in team memo
- ❌ Appendix is generic with no details

Note: Even if memo is perfect, **missing AI Audit = 0/50 points** per course policy.

Bonus Points (Optional, up to +5 points)

Exceptional work beyond requirements:

- **+2 points:** Comprehensive scenario analysis (3+ scenarios with probability-weighted expected returns)
- **+2 points:** Risk dashboard (VaR, stress tests, or Monte Carlo simulations)
- **+1 point:** External validation (compare model predictions to actual 2024 REIT performance)

- **+1 point:** Interactive visualizations (e.g., Plotly dashboard embedded in memo PDF)

Total Possible: 55 points (50 base + 5 bonus)

Example Grade Calculations

Team A: Excellent Work (48/50 + 2 bonus = 50/50)

Team Memo (40 points):

- Reproducibility & Rigor: 10/10 (code perfect, all diagnostics run)
- Structure & Clarity: 10/10 (all sections, professional, 6 pages)
- Results & Interpretation: 12/12 (tables publication-ready, interpretation excellent)
- Recommendations & Caveats: 8/8 (specific recommendations, honest limitations)

Individual Addendum (10 points): (Average across 4 students)

- Student 1: 10/10 (excellent contributions, defended decision, key limitation)
- Student 2: 9/10 (strong work, minor gap in defended decision)
- Student 3: 10/10 (excellent)
- Student 4: 9/10 (strong work)
- **Average:** 9.5/10 (rounded to 10)

Bonus: +2 (comprehensive scenario analysis with 5 scenarios)

Final Grade: 50/50 (capped at 50)

Team B: Satisfactory Work (37/50)

Team Memo (30 points):

- Reproducibility & Rigor: 7/10 (code runs, but 1-2 diagnostics missing)
- Structure & Clarity: 8/10 (all sections, but some jargon remains)
- Results & Interpretation: 9/12 (tables good, interpretation solid but not deep)
- Recommendations & Caveats: 6/8 (recommendations vague, limitations mentioned but not deep)

Individual Addendum (7 points): (Average)

- Students provided adequate contributions and defended decisions, but not exceptionally detailed

Final Grade: 37/50 (B)

Team C: Needs Improvement (24/50)

Team Memo (18 points):

- Reproducibility & Rigor: 5/10 (code has errors, diagnostics incomplete)
- Structure & Clarity: 5/10 (missing 1 section, heavy jargon)
- Results & Interpretation: 6/12 (tables are raw output, interpretation minimal)
- Recommendations & Caveats: 2/8 (no specific recommendations, no limitations)

Individual Addendum (6 points): (Average)

- Contributions vague, defended decisions weak

Final Grade: 24/50 (D+)

Team D: Unsatisfactory (0/50)

- Missing AI Audit Appendix in team memo
- **Grade:** 0/50 (automatic zero per course policy)

Rubric Summary Table

Component	Excellent	Satisfactory	Needs Improvement	Unsatisfactory
Team: Reproducibility (10)	9-10	6-8	3-5	0-2
Team: Structure (10)	9-10	6-8	3-5	0-2
Team: Results (12)	11-12	8-10	5-7	0-4
Team: Recommendations (8)	7-8	5-6	2-4	0-1
Individual: Contribution (3)	3	2	1	0
Individual: Decision (3)	3	2	1	0
Individual: Limitation (3)	3	2	1	0
Individual: Quality (1)	1	-	-	0
Total (50)	45-50	32-44	17-31	0-16
AI Audit	Pass (required)	Pass (required)	Pass (required)	Fail = 0/50
Bonus (optional)	+0 to +5	-	-	-

Grading Timeline

- **Submission Deadline:** Friday, Week 14 (May 1) by 11:59 PM (same calendar date as Final Presentations Day 1; see `QM2023-Spring2026-Week-Date-Map.md`)
- **Instructor Grading:** Completed within 1 week (by May 11)
- **Final Grades Posted:** Blackboard gradebook + memo comments on GitHub

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How to Use This Rubric

For Students (Self-Assessment Before Submission)

Team Memo Checklist:

- ☐ All sections present (Executive Summary, Methodology, Results, Conclusions, References, AI Audit)
- ☐ Length is 5-7 pages
- ☐ Tables are publication-ready (not raw Python output)
- ☐ Figures are high-resolution (300 DPI) with titles, labels, legends
- ☐ Investment recommendations are specific (not vague)
- ☐ Limitations are honest and substantive
- ☐ No jargon or unexplained acronyms
- ☐ Professional tone, no typos

Individual Addendum Checklist:

- ☐ Contributions are specific (tasks + hours for each milestone)
- ☐ Defended decision is clear and grounded in evidence
- ☐ Key limitation is substantive (not trivial)
- ☐ Exactly 1 page (not 0.5 or 2 pages)
- ☐ Professional writing, no typos

For Instructors (Grading Guide)

1. **Team memo first:** Assess reproducibility, structure, results, recommendations
2. **Individual addenda:** Grade each student separately on contribution, defended decision, limitation
3. **AI Audit check:** Ensure appendix is present and detailed (if missing → 0/50 for team)
4. **Feedback:** Provide specific, constructive comments

Example Feedback:

> *"Your memo is well-structured and professional (10/10 Structure). However, your investment recommendations are vague ('Be cautious') instead of specific ('Overweight Industrial by 15%'). Also, you didn't acknowledge the parallel trends assumption in your DiD analysis—this is a critical limitation. (-2 Recommendations, -2 Caveats). Total: 46/50 (A-). Strong work overall; refine recommendations for future memos."*

Frequently Asked Questions

Q: Can we use bullet points in the Executive Summary?

A: No. Executive Summary should be 2-3 paragraphs of prose (no tables, figures, or bullet lists). It's designed to be read by busy executives who want a quick narrative summary.

Q: Our memo is 8 pages. Will we lose points?

A: Likely yes. If it's 8 pages with essential content, minor penalty (1-2 points). If it's 8 pages with fluff, more significant penalty (3-5 points). **Target: 5-7 pages.**

Q: We disagree on investment recommendations. How do we handle this?

A: Present the recommendation the team agreed on by majority vote. If disagreement is genuine, you can note it: "Our team considered both overweighting and neutral stance on Retail REITs. We chose neutral due to [X], but acknowledge that an overweight strategy could be defensible given [Y]."

Individual Addendum: If you personally disagree, you can note this in your "Defended Decision" section and explain your alternative reasoning.

Q: Can we include our code in an appendix?

A: No. The memo is for the Investment Committee (non-technical). Code should be in your GitHub repo, not in the PDF memo. **Exception:** Short code snippets (2-3 lines) in footnotes to explain a methodological choice, but this is rare.

Q: What if we ran out of time and didn't polish the memo?

A: You'll lose points for Structure & Clarity (typos, formatting errors) and potentially Recommendations (if rushed and vague). **Time management is part of the assessment.** Start the memo during the Week 14 hackathon; submit by **11:59 PM Friday, May 1** (end of Week 14).

Grading Philosophy

This rubric emphasizes:

1. **Professional communication:** Can you translate technical analysis into actionable business advice?
2. **Intellectual honesty:** Do you acknowledge limitations and uncertainty?
3. **Evidence-based recommendations:** Are your investment suggestions grounded in empirical findings?
4. **Individual accountability:** Did you contribute meaningfully and reflect thoughtfully?

We are not grading:

- Technical sophistication of models (that was M3)
- Length of memo (concise is better than verbose)
- Creative visualizations (functional is fine)

We are grading:

- Can a non-economist understand your findings?
- Are your recommendations specific and defensible?
- Did you honestly assess strengths and limitations?

Rubric prepared for QM 2023: Statistics II, Spring 2026, University of Tulsa Contact: Dr. Cayman Seagraves (cayman-seagraves@utulsa.edu)